24/09/2026

Exp no: 07 Sliding Window Protocol

Aim: Write a program to implement flow control at data link layer using sliding window protocol.

Program:

```
import time
message = input("Enter the text message: ")
window_size = int(input("Enter the window size: "))
error_frame = int(input("Enter frame: "))
if window_size <= 0:
    print("Invalid window size")
    exit()
print("SLIDING WINDOW PROTOCOL")
print("Window size:", window_size)
start = 0
expected_frame = 1
error_induced = False
while start < len(message):
    end = min(start + window_size, len(message))
    print("SENDER BUFFER")
    sender_buffer = []
    for i in range(start, end):
        frame_no = i+1
        data = message[i]
        sender_buffer.append([frame_no, data])
        print("Frame_no =", frame_no, "Data =", data)
print("\n Sending frames...")
time.sleep(1)
print("RECEIVER")
error_found = False
for frame in sender_buffer:
    frame_no = frame[0]
    data = frame[1]
```

```
if frame_no == error_frame and not error_induced:
    received_frame = frame_no + 1
    error_induced = True
else:
    received_frame = frame_no
print("Recived Frame: ", received_frame,
      "DATA:", data)
if received_frame == expected_frame:
    print("Frame ", received_frame, " is correct")
    expected_frame += 1
else:
    print("\n ERROR DETECTED")
    print("expected Frame: ", expected_frame)
    print("Recived frame: ", received_frame)
print(" RECEIVER BUFFER")
print("NACK = ", expected_frame)
print("\n NACK received by sender ")
print("Resending frames from:", expected_frame)
error_found = true.
break
if error_found:
    start = expected_frame + 1
    time.sleep(1)
    continue
print("RECEIVER BUFFER")
ack = expected_frame
print("Ack = ", ack)
print("\n ACK received by sender ")
print("Expected ACK: ", end + 1)
if ack == end + 1:
    print("\n ACK is correct ")
    print(" sliding window moves forward ")
    start = end
```

```
else:
    print("\n Incorrect ACK")
    print("Waiting for correct ACK")
print("Transmission completed")
print("\n Original Message :", message)
print("All frames successfully transmitted")
```

Output:

```
Enter the test message: hello
Enter the window size : 3
        SLIDING WINDOW PROTOCOL
Message : hello
Window size : 3
        SENDER BUFFER
Frame-no = 1    Data = h
Frame-no = 2    Data = e
Frame-no = 3    Data = l
Sending frames...
        RECEIVER
Received frame : 1  Data : h
Frame 1 is correct
Received frame : 2  Data : e
Frame 2 is correct
Received frame : 8  Data : l
        ERROR DETECTED
Expected frame : 3
Received frame : 8
        RECEIVER BUFFER
NACK = 3
NACK received by Sender
Resending frames from frame 3.
```

SENDER BUFFER

Frame-no = 3 Data = l
Frame-no = 4 Data = l
Frame-no = 5 Data = O

Sending frames ...

RECEIVER

Received Frame : 3 Data : l
Frame 3 is correct
Received Frame : 4 Data : l
Frame 4 is correct
Received Frame : 5 Data : O
Frame : 5 is correct

RECEIVER BUFFER

Ack = 6
Ack. received by sender
Expected Ack = 6
Ack is correct
Sliding window move forward

Transmission Completed
Original Message : hello
All frames successfully transmitted.

Result : Thus flow control at data link using sliding window protocol is done successfully.